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19 August 1980
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MILITARY SPECIFICATION

# COMPRESSOR UNIT, RECIPROCATING, POWER-DRIVEN, AIR, 15 CFM, 3,500-PSI

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers 15-cfm, 3,500-psi, power-driven, portable air compressors.

1.2 Classification. Compressors shall be of the following types and classes, as specified (see 6.2):

Type I - Skid-mounted.
Type II - Wheel-mounted.

Class 1 - Gasoline-engine-driven.
Class 2 - Electric-motor-driven.

## 2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 4310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

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| CC-M-1807 | - Motors, Alternating Current, Fractional and Integral Horsepower, (500 HP and Smaller). |

MILITARY

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| MIL-V-173 | - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment). |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| MIL-T-704 | - Treatment and Painting of Materiel. |
| MIL-L-2105 | - Lubricating Oil, Gear, Multi-purpose. |
| MIL-H-3177 | - Heaters, Coolant, Engine: Multifuel-Burning, 24-Volt D.C. |
| MIL-H-3199 | - Heaters, Vehicular Compartment 24 Volt, D.C. |
| MIL-F-3541 | - Fittings, Lubrication. |
| MIL-C-3600 | - Compressors, Rotary, Power-Driven; and Compressors, Reciprocative, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of. |
| MIL-G-3859 | - Grease Guns, Hand-Operated, Lever, Push and Screw Type. |
| MIL-F-5504 | - Filters and Filter Elements, Fluid Pressure, Hydraulic Micronic Type. |
| MIL-H-7365 | - Hose, Air Duct, for Ground Heaters. |
| MIL-R-8573 | - Reservoirs, Air, Nonshatterable Steel. |
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-B-11040 | - Belt, V: Engine Accessory Drive. |
| MIL-G-23827 | - Grease, Aircraft and Instrument, Gear and Actuator Screw. |
| MIL-T-25363 | - Tank, Pneumatic Pressure, Aircraft, Glass Fiber. |
| MIL-C-26058 | - Cartridge, Gas Purifier, Extremely Low Dewpoint, Type MA-2. |
| MIL-H-26666 | - Hose Assembly, Pneumatic, High Pressure. |
| MIL-H-38390 | - Hose Assembly, Tetrafluoroethylene, Peneumatic, High Pressure. |



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| MIL-H-52471/2 | - Hose Assemblies, Rubber: Hydraulic Pressure-Type; Type 100R2 |
| MIL-C-83959 | - Cylinder, Air Purifier. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-H28 | - Screw-Threads Standards for Federal Services |

MILITARY

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| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-209 | - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment. |
| MIL-STD-461 | - Electromagnetic Interference Characteristics Requirements for Equipment. |
| MIL-STD-642 | - Identification Marking of Combat and Tactical Transport Vehicles. |
| MIL-STD-889 | - Dissimilar Metals. |
| MIL-STD-1410 | - Methods for Selection of Industrial Engines for End Item Application. |
| MS28889 | - Valve, Air, High Pressure Charging. |
| MS51331 | - Guards, Splash, Wheel. |
| MS51335 | - Pintle Assembly, Towing, 18,000 Lbs. Capacity, Manual Release. |
| MS51338 | - Coupler, Drawbar, Ring - Light Duty. |
| MS53028 | - Chassis, Trailer, 3/4 Ton, 2-Wheel, Military Design, M116A1. |
| MS53063 | - Indicator, Air Cleaner, Intake; Restriction Mechanical Type. |

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Performance Test Code for Displacement Compressors, Vacuum Pumps, and Blowers, PTC 9.
Boiler and Pressure Vessel Code, Section VIII, Unfired Pressure Vessels.



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(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

RUBBER MANUFACTURERS ASSOCIATION

Engineering Standards for Multiple V-Belt Drive.

(Application for copies should be addressed to the Rubber Manufacturers Association, 444 Madison Avenue, New York, NY 10022.)

NATIONAL FIRE PROTECTION ASSOCIATON

NFPA No. 70 - National Electrical Code (Electrical Wiring and Apparatus).

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, New York, NY 02110.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

FOREST SERVICE

Forest Service Standard 5100-1.

(Application for copies should be addressed to the Forest Service, US Dept of Agriculture, P.O. Box 2417, Washington, DC 20013.)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29CFR Part 1910, 212a

(Application for copies should be addressed to Bureau of National Affairs, 1231 25th Street NW, Washington DC 20037.)

ASSOCIATION OF AMERICAN RAILROADS

Loading and Tiedown Requirements



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(Application for copies should be addressed to the Association of American Railroads, 1920 L Street NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. Each compressor unit (herein referred to as compressor) shall consist of the following major components:

(a) Compressor pump.
(b) Power unit (gasoline engine for Class 1 and electric motor for Class 2).
(c) Dehydrator unit.
(d) Moisture separator.
(e) Winterization system, when specified (see 3.9.12).
(f) Enclosure.
(g) Mounting base (skid-mounted for Type I and wheel-mounted for Type II).
(h) Coolers, traps, servicing line, and air and oil filters.
(i) Receiver.

3.2 First article (preproduction model). The contractor shall furnish one or more compressors for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce compressors that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests.

3.3 Initial production compressor. When specified (see 6.2), the contractor shall furnish to the Government one or more compressors for inspection as specified in 4.4.

3.4 Material. Material shall be specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9).

3.4.1 Dissimilar metals. There shall be no contact between dissimilar metals which could cause deterioration, galvanic corrosion, seizure, or warpage through differential contraction or expansion. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-STD-889 shall be insulated as specified in MIL-STD-889.

3.4.2 Corrosion-resisting treatment. Corrosion-resisting treatment shall consist of galvanizing or electrodepositing of chromium, copper, or nickel.



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3.5 Environmental requirements.

3.5.1 Operation at extreme temperatures. The compressor shall operate as specified herein under the following conditions.

3.5.1.1 Low temperatures. The nonwinterized compressor shall start and operate as specified herein for not less than 2 hours at rated capacity in ambient temperature at minus 25° F. With a winterization system installed and operating, the compressor shall start and operate as specified herein for not less than 2 hours at rated capacity in ambient temperature minus 50° F (see 3.9.12).

3.5.1.2 High temperatures. The compressor, either nonwinterized or winterized, shall start and operate as specified herein at any ambient temperature up to plus 120° F for not less than 4 hours at rated capacity.

3.5.2 Rain. The compressor in operating condition shall start and operate as specified herein after exposure to 1/4 hour of rainfall at a minimum rate of 3 inches per hour directed at all four sides and top. The compressor shall operate as specified herein for 1/4 hour at no load while the rain continues to fall at the minumum rate of 3 inches per hour.

3.6 Safety and human factors.

3.6.1 Safety. All parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded when such parts are exposed to contact by personnel, or otherwise create a hazard. All moving parts which are hazardous to personnel shall be fully enclosed or guarded in accordance with OSHA regulations. Exhaust or discharges from the compressor shall be directed so that they do not endanger personnel. Protective devices shall not impair the operating functions. Sharp edges, projecting points, shall be avoided. On fastening devices, protruding lengths of threads shall be at least two full threads but no more than six full threads. Warnings shall be mounted on or near components containing hidden hazards such as automatic blow-down discharges.

3.6.2 Equipment. Safety devices in accordance with MIL-STD-1410 shall be provided to automatically stop the engine when any of the limiting temperatures or pressures recommended by the engine manufacturer are reached. Switches of the mercury type and those employing a gage needle to activate the safety device shall not be used.

3.6.3 Human factors. The characteristics of the compressor shall provide for operation and maintenance by personnel ranging from the small man (5th percentile), clothed, through the large man (95th percentile), with arctic clothes, when applicable, in accordance with SAE J833.

3.6.4 Noise. The noise level at the operator position at the control panel while the compressor is delivering rated capacity should be within the limits of 85 dB(A). If the noise level is not within these limits, the distance from the control panel at which 85 dB(A) is met shall be determined. That distance shall be shown on a warning plate which shall be mounted on both sides of the



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compressor, starting "Warning ear protection required within ____ feet when equipment is operating" (see 4.6.2.12). The warning shall be legible from the distance where the levels are met. In no instance shall the distance for compliance be greater than 15 feet.

3.7 Maintainability and reliability.

3.7.1 Maintainability. The compressor shall operate as specified herein without repair or maintenance other than the contractor's recommended normal scheduled adjustments and servicing, as established by a maintenance schedule prepared and submitted by the contractor prior to test. All major assemblies shall be accessible for servicing, repair, and replacement without the removal of other major assemblies, utilizing conventional tools associated with items of this nature. Cover plates which must be removed for component adjustment or repair shall be equiped with quick-disconnect fastenings. All fasteners shall be of corrosion-resistant materials or shall be treated to be corrosion resistant. All screw threads shall be in accordance with FED-STD-H28. Maximum use shall be made of interchangeable hardware and fastening devices. Use of self-tapping screws shall not be permitted. Means for drainage of lubricants, fuel, and condensate traps shall be in an accessible location, and shall drain to a container without splashing on any component or on operating personnel. Integral tubes, pipes, or other means shall be used to convey draining fluids to a container. The drain outlets shall be so located to facilitate complete drainage. Each maintenance assembly or disassembly operation performed as a result of testing in accordance with 4.6.2.15.3 shall be accomplished by no more than two men using common tools and special tools furnished with the compressor. The ratio of the man-hours of maintenance required to the hours of operation performed shall not exceed 0.06.

3.7.2 Ease of maintenance. Ease-of-maintenance provisions shall incorporate, to the maximum practicable extent, features insuring operating clearances and facilitating applicable maintnance and servicing operations at low temperatures by personnel wearing heavy mitten-type gloves and handicapped by bulky clothing and footgear. Devices and controls with threaded fastenings which can be over-torqued by operators lacking feel through gloves shall be avoided (see 3.6.3).

3.7.3 Reliability. The "specified" Mean-Time-Between-Failure (MTBF) shall be 216 hours when the compressor is tested as specified in 4.6.2.15.2.

3.8 Performance.

3.8.1 Rated capacity. Within one hour of operation after starting, the compressor shall deliver not less than 15 cubic feet per minute (cfm) of free air at a delivery pressure of 3,500 pounds per square inch (psi) (see 6.6.1 and 6.6.3) when operating under all conditions specified herein. The moisture content of the delivered air shall be no greater than 23.6 parts per million (ppm) by volume (minus 65° F dewpoint at 14.7 pounds per square inch obsolute (psia)).

3.8.2 Endurance. The compressor shall operate as specified herein for 500 hours when tested in accordance with 4.6.2.4. At the end of this 500 hour



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period, the decrease in output shall be not more than 1.5 cfm when delivering air at a pressure of 3,500 psi.

3.8.3 Tilted operation. The compressor shall deliver the rated capacity at 3,500 psi for not less than 30 minutes when tilted in any position at an angle of 15 degrees from the horizontal with the oil level in the engine and compressors on the "low" or "add" mark when checked with compressor unit sitting on a level surface. The compressor shall stand for not less than 4 hours in a tilted position without any evidence of leaks; faulty lubrication, cooling, or fuel supply and distribution; lubricant spill-over; or reduced performance upon starting after standing in this position.

3.9 Compression system.

3.9.1 Air compressor pump. The air compressor pump shall be interchangeable as an assembly for both type I and type II compressors and for both class 1 and class 2 compressors when procured from a single contractor.

3.9.2 Cooling. The compressor shall be entirely air cooled. Intercoolers shall be provided to remove the heat of compression between consecutive stages. An aftercooler shall also be provided. All coolers shall have the capacity to cool the air after compression to a temperature of not more than the ambient temperature plus 50° F in the ambient temperature of 120° F. In the ambient temperature range of plus 30° F to minus 50° F, positive control of cooling shall be provided to prevent valves, traps, piping, and controls from freezing. If a manual device is used, a means shall be provided on the control panel to indicate to the operator the correct positioning of the device. Each cooler shall be provided with an adjustable, spring-loaded-type, pressure-relief valve located upstream from each cooler, adjusted to operate at the pressure necessary to protect the compressor. On winterized compressors, the compressor cooling airflow shall be controllable so as to permit efficient operation at an ambient temperature of minus 50° F.

3.9.3 Pressure vessels. All pressure vessels shall be designed for minimum operating pressure of 3,500 psi and a proof pressure of 5,250 psi and shall conform to the requirements of the ASME Code for Unfired Pressure Vessels unless otherwise specified herein.

3.9.4 Pressure tubing and fittings. All pressure tubing and fittings shall have a minimum operating pressure of 3,500 psi and a proof pressure of 5,250 psi, and shall be of corrosion-resistant steel containing a minimum of 18 percent chromium and 8 percent nickel. All tubing and fittings shall be installed to minimize and withstand vibration and allow for thermal expansion. Provision shall be made for relieving pressure to the atmosphere from all lines in the system without disassembly of any fitting or component.

3.9.5 Valves. Manually operated shutoff, insolation, and control service valves shall be of the positive shutoff type and shall be of corrosion-resistant steel containing a minimum of 18 percent chromium and 8 percent nickel. The valves shall require not more than 60 inch-pounds of torque to open or close



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under a pressure of 3,500 psi and over-torquing to 150 inch-pounds shall not damage the valve seats. Valve seats shall be resistant to seat erosion.

3.9.6 Moisture separator. A mechanical-type moisture separator shall be provided downstream from the aftercooler to remove condensed moisture and oil. The separator shall have a minimum operating pressure of 3,500 psi and a proof pressure of 5,250 psi, and shall be of a corrosion-resistant material. The separator shall have a volumetric capacity large enough to contain the moisture and oil which would accumulate in 20 minutes while the compressor is operating at rated capacity with a relative humidity of not less than 85 percent and an ambient temperature of 120° F. A valve shall be provided for periodic drainage of the moisture separator. Periodic drainage shall not be required more frequently than 15 minute intervals under all conditions specified herein. In addition, automatic drainage of the moisture separator shall be provided on shutdown of the compressor. Provision shall be made to prevent freezing of moisture when operating at ambient temperature at ambient temperatures below 32° F. When an electrical heater blanket is utilized, it shall be actuated by a manual switch. Provisions shall be made to prevent actuation of the heater when the unit is not is use.

3.9.7 Dehydrator. The chemical-type dehydrators shall be provided and shall be of the replaceable cartridge type, conforming to MIL-C-83959 and provided with an MA-2 cartridge conforming to MIL-C-26058. Provisions shall be made for relieving pressure to the atmosphere.

3.9.7.1 Mounting. The cartridge holder shall be rigidly mounted in an accessible location and in a vertical position. The cartridge holder shall be connected so that the air leaving the receiver will enter the first cartridge holder at the bottom, leave at the upper connection, enter the second cartridge holder at the bottom connection, and leave at the top.

3.9.7.2 Operating instructions. Instructions for changing the cartridges in the cartridge holders shall be provided on an instruction plate (see 3.16). Instructions shall also include the required hours between cartridge changes for not less than three temperature ranges to maintain a delivery air dewpoint of minus 65° F as determined during preproduction testing. Interpolation and extrapolation of test results will be permitted. A blank space shall be provided on the instructon plate on which the hourmeter reading (see 3.9.19.2) can be recorded and erased. The cartridge holders shall be labeled No. 1 (cartridge holder following receiver) and No. 2 (outlet cartridge holder). Change cartridges as follows:

(a) Discard cartridge in No. 1 (inlet) holder.
(b) Transfer cartridge from No. 2 (outlet) holder to No. 1 holder.
(c) Place fresh cartridge in No. 2 holder.

3.9.7.3 Pressure control valve. An automatic 1/4 inch pressure control valve shall be provided to prevent the passage of air through the moisture separator or dehydrator at pressures less than 2,000 psi. A check valve shall be provided to prevent reverse airflow.



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3.9.8 Air filter.

3.9.8.1 Intake air filter. A dry type air filter shall be furnished for air being inducted into the first stage of the air compressor. Each air filter shall be provided with an air indicator conforming to MS53063-3 which shall be either located at the air cleaner or the control panel (see 3.9.19) in a position visible to the operator.

3.9.8.2 Moisture separator inlet air filter. An accessible and easily serviced filter shall be provided at the inlet to the moisture separator to avert any foreign matter from entering the moisture separator.

3.9.8.3 Discharge air filter. The compressed-air discharge shall be provided with a filter which shall have a nominal rating of 10 microns and an absolute rating of 25 microns conforming to MIL-F-5504. The filter element shall not impair the ability of the compressor to meet minimum performance requirements for at least 50 hours before filter change. The filter shall be in an accessible location so that removal and replacement of the filter element shall not require more than 10 minutes

3.9.9 Pressure relief valves. Adjustable, preset, spring-loaded pressure-relief valves shall be provided to protect the moisture separator, air receiver, and the individual stages of the air compressor. The adjustment, once made, shall be securely locked and sealed to prevent unauthorized adjustment. The last stage of the compressor pump may be protected by the use of a rupture disk. In addition, rupture disks shall be provided to protect the moisture separator and air receiver. The relief valves and rupture disks shall be capable of discharging 100 percent of the compressor capacity.

3.9.10 Receiver. A receiver shall be provided upstream of the dehydrator. The receiver shall withstand an operating pressure of 3,500 psi and, unless otherwise specified (see 6.2), shall have a volume capacity of not less than 1300 cubic inches. The receiver shall be equipped with a drain valve located at the lowest point of the receiver, a spring-loading-type pressure-relief valve set at 4,000 psi plus or minus 100 psi, and a disk-type pressure-relief valve set for a range of 4,400 to 4,800 psi. The construction shall conform to the Code for Unfired Pressure Vessels of the ASME Boiler and Pressure Vessel Code. The receiver shall bear the ASME stamp, and no welding or other activation shall be permitted after the receiver has been inspected. When a fiber glass receiver is utilized, it shall conform to MIL-T-25363 and to the requirements of this specification. A pressure vessel conforming to MIL-R-8573 is also acceptable provided it meets the requirements of this specification.

3.9.10.1 Unloader. A device shall be provided to unload the compressor automatically when the discharge pressure reaches 3,500 psi plus 100 psi minus 0 psi, and to load the compressor at a pressure between 3,000 psi and 3,500 psi.

3.9.10.2 Blowdown device. An automatic blowdown device shall be provided to discharge the air receiver and entire compression system when the engine stops. The device shall reduce the air receiver pressure to not more than 10 psi.



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3.9.11 Enclosure. A protective housing, constructed of steel of 0.048 inch nominal thickness or of aluminum of 0.064 inch nominal thickness, shall be furnished. A metal pocket for technical publications shall be provided on the inside of the enclosure. The housing shall completely enclose the sides, ends, bottom, and top of the compressor. The housing shall be provided with tight hinged doors and hatch covers which hold tightly and securely when closed against extremely adverse and severe weather conditions. The handles of the hinged doors or hatch covers shall not extend beyond the outer sides of the end item. The handles shall be operable by personnel wearing heavy mitten-type gloves. The enclosure shall prevent the entry of wind, water, dust, and snow. Drip channels or other similar facilities shall be provided to prevent entry of snow or rain. The enclosure shall not be welded to the frame and shall be capable of installation and removal without special tools. Provisions shall be made for the entry of cooling air into the interior of the compressor. The enclosure shall not be used to support heavy accessories, such as battery boxes and gasoline tanks.

3.9.12 Winterization. When specified (see 6.2), a winterization system shall be provided to maintain a temperature of at least 0° F inside the enclosure at an outside ambient temperature of minus 50° F when the power unit is not running. The heater on the class 1 compressor shall be complete with its fuel control and combustion system. The fuel for the heater shall be supplied from the fuel tank which supplies the engine. The heater shall conform (except for capacity) to the functional requirements of MIL-H-3177, capacity optional, or MIL-H-3199, type I, capacity optional and it shall be designed for a 12 volt system. The heat shall be distributed to the engine and compressor crankcases and the battery by means of ducts, blowers, and heat controls. Exhaust gases shall not be vented inside the enclosure. On the class 2 compressor, an immersion-type heater shall be furnished. The heater shall maintain oil at operating temperatures during standby, and shall attain oil operating temperatures in 60 minutes after being cold soaked to minus 50° F.

3.9.12.1 Heat duct attachments. Provisions shall be made on winterized compressors for the attachment of either one 12-inch-diameter duct or three 6-inch-diameter ducts, both in accordance with MIL-H-7365, for use in directing warm air inside the enclosure. Covers for duct openings shall be provided if necessary. Removable duct opening covers shall be secured to the compressor by chains or other similar means to prevent their loss. Duct attachments and covers shall not project beyond the enclosure of the compressor.

3.9.12.2 Oil dilution. Oil dilution for starting and running the compressor at low outside ambient temperature shall not be accepable (see 3.9.22.1).

3.9.13 Moisture traps. Traps shall be mounted so as not to be affected by any extremes of temperature specified herein.

3.9.14 Sensitive elements. Sensitive elements such as fuel, cooling, and lubrication systems, storage batteries, pressure gages, drain valves, and connecting tubing shall be so arranged within the enclosure as to prevent exposure to extremes of temperature.



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3.9.15 Drain valves. Drain valves shall be grouped and mounted on a common member. Drain valve handles shall not extend beyond the exterior of the enclosure, shall be cool enough to be operable with bare hands, and shall be operable by personnel wearing heavy mitten-type gloves.

3.9.16 Drive belts. Drive belts shall conform dimensionally to RMA Engineering Standards for Multiple V-Belt Drives and shall conform to the material requirements of MIL-B-11040. A means for adjusting belt tension shall be provided.

3.9.17 Hose assemblies. Hose for hydraulic, oil, and coolant lines, excluding radiator hose assemblies, shall conform to MIL-H-52471/2. Hose assemblies for pneumatic lines shall conform to MIL-H-26666 or MIL-H-38390, -6 size.

3.9.18 Servicing hose. A 30-foot pneumatic hose assembly of the size specified in 3.9.17 shall be provided. The hose shall be provided with a quick-disconnect fitting for an operating pressure of 3,500 psi and mating with the terminal fitting of the high pressure servicing system mounted on the control panel (see 3.9.19). The coupling shall be a two-way shutoff and shall be a Hansen series 2 HK or equal. It shall be constructed of corrosion-resistant steel. An air chuck which shall fit valve assemblies conforming to MS28889 shall be provided on the servicing end of the pneumatic hose assembly. The hose shall be stowed in an accessible place inside the enclosure. The hose shall be secured in a manner to prevent fraying caused by vibration during preparation for use, during usage, and during stowage. Two servicing valves shall be provided; one located between the servicing line and the air chuck, and the other located between the dehydrator outlet and the servicing line to control the flow of air into the servicing line. A manually operated value for bleeding shall be provided to relieve the pressure in the servicing line prior to disconnecting. A spring-loaded valve shall be provided between the bleed valve and the atmosphere to prevent dirt and moisture from entering the hose assembly when pressure is relieved.

3.9.19 Control panel. A control panel shall be provided for mounting the gages, meters, and controls as specified herein. Gages, meters, and controls shall be clearly identified (see 3.16) and shall be mounted so that they will not be damaged under any condition of compressor operation specified herein. All controls, gages, and meters necessary for operation shall be accessible and visible to a single operator standing on one side of the compressor. A schematic airflow diagram shall be permanently affixed on or near the control panel.

3.9.19.1 Instrumentation. Pressure gages shall be provided to indicate intercooler, aftercooler, receiver, and servicing line pressures. All pressure gages shall be mounted on the control panel, shall be isolated from vibration by shock-absorbing mounts and, where applicable, shall be connected to the hot end of the coolers. Each gage shall be the liquid fill type to control needle vibration. Provision shall be made to prevent freezing of gages under any of the temperature conditions specified herein.

3.9.19.2 Hourmeter. An hourmeter shall be mounted on the control panel. The hourmeter shall have a range of from 0 to 9,999 and an accuracy of 1 minute per



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hour. The hourmeter may be of the revolution-counter type. The hourmeter shall register only when the compressor is running.

3.9.20 Lighting. Electric lighting for the control panel and for the compressor compartment shall be provided. Electric lights shall be provided for blackout operation of the control panel. Incandescent lamps shall be shielded so that the bare lamp is not visible from aerial or ground observation. Arrangement of the blackout lamps shall be such that all instruments, control knobs, and switches are identifiable and readable. The type to be used shall have an adjustable lens cover which will vary the light from off to full intensity.

3.9.21 Electromagnetic interference. The electromagnetic interference emission and susceptability characteristics of the compressor shall conform to MIL-STD-461, Tactical and Special Purpose Vehicles and Engine Driven Equipment.

3.9.22 Lubricaton. All surface requiring lubrication shall be provided with a means for lubricating. A dipstick oil-level indicator, a crankcase ventilator, an oil drain, and a pressure switch shall be provided. When the crankcase ventilator is open to the atmosphere, a filter shall be provided to prevent foreign matter from entering the crankcase. The oil drain shall be not less than 3/4-inch diameter and shall be located to permit complete drainage of the sump. The pressure switch shall be provided on pressurized lubrication systems to stop the compressor in event of compressor oil-pressure failure.

3.9.22.1 Lubricants. The compressor shall operate as specified herein when lubricated with the following military lubricants:

(a) Oil conforming to and qualified under a military specification (see 6.4).
(b) Grease conforming to MIL-G-10924 or MIL-G-23827.
(c) Universal gear lubricant conforming to MIL-L-2105.

3.9.22.2 Lubrication fittings. Lubricaion fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859. Accessibillity to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasters may be removed to provide accessibility to lubrication fittings.

3.9.22.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.9.22.4 Filling, draining, and checking provisions. Enclosures that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be located so that removal of the plug will result in complete



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drainage of the lubricant from the enclosure. Drainage shall be to a container when the compressor is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to a container. Accessibility to the drainplug, the filling means, and the lubricant level checking device shall be provided without the removal or adjustment of accessories or parts. Compressor housing side panels or plates equipped with hand-operable, quick-disconnect fasteners may be removed.

3.9.22.5 Delivery lubrication service. When delivered, all components, except those for which preservatives are specified in 5.1, shall be lubricated. Grease lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range as specified herein. Each lubricated component shall be tagged in a conspicous place to indicate the temperature range and grades of oils and greases used.

3.10 Fungus and moisture resistance. The electrical circuitary, including all components and connections except as specified below, shall be protected from the effects of fungus growth and mositure by an overall treatment with varnish conforming to MIL-V-173, composition as specified (see 6.2), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.10.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.11 Lifting attachments. Permanently affixed attachments that enable the compressor to be lifted in its normal travel or operating position shall be provided and shall comply with MIL-STD-209, type IV. Lifting attachments may be also be used as tiedown devices when such attachments meet the requirements specified in 3.12.

3.12 Tiedown attachments. Permanently affixed attachments that permit fastening the compressor to the floor or deck of a transportation medium shall



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be provided and shall comply with MIL-STD-209, type IV. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.11.

3.13 Transportability by rail. The compressor shall withstand the impact forces encountered in rail shipment as tested in 4.6.2.10 without damage or deformation to any part of the compressor.

3.14 Identificaton marking. The compressor shall be identified in accordance with MIL-STD-130. The marking shall be applied to the compressor on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.15 Shipping data plate. Shipping data plates shall be provided and shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and in additon, shall show the silhouette of the compressor in transport position indicating the center of gravity and the location and capacity of the lifting and tiedown attachments. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.16 Instruction plates. Each compressor shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the compressor. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.17 Treatment and painting. The portions of the compressor and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, Lusterless No. 34087.

3.18 Stenciling. The gross weight shall be stenciled on each side of the compressor. Stenciling shall conform to MIL-T-704 using block or stencil-type letters not less than 1 inch high. On type II compressors, the registration numbers and other marking of the compressor shall be in accordance with MIL-STD-642. The gross weight shall be marked on each forward side of the compressor. The tire pressure shall be marked on the side of the trailer near the tires.

3.19 Toolbox. The compressor shall be provided with a toolbox made of 0.075-inch nominal thickness metal. The toolbox shall have an over-lapping hinged lid which prevents the entry of rain and dirt. The lid shall be provided with a fastener of a type that will keep the lid closed when the toolbox is subjected to vibration. The toolbox shall be fastened to the compressor in a protected position. The toolbox shall be of sufficient size to house the operating and maintenance tools and shall incorporate a provision for drainage.

3.20 Government-loaned property. For type I compressors (see 6.5), the Government will loan a military, 3/4-ton, 2-wheel, M116 trailer chassis



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conforming to MS53028, to aid the contractor in designing the skid base for mounting on the M116 trailer chassis.

3.21 Type I, skid-mounted.

3.21.1 Dimensions. The overall outside dimensions of the type I compressor, including the skid, shall be not more than 72 inches long by 60 inches wide by 46 inches high.

3.21.2 Weight. The weight of the type I compressor, including the skid, shall be not more than 2,100 pounds.

3.21.3 Fenders. When specified (see 6.2), the contractor shall furnish two fender assemblies. Each fender assembly shall consist of a fender and splash guard conforming to SAE J321, SAE J682, and SAE J683, and necessary mounting brackets, which shall restrict water and other road-surface substances thrown from the wheels. Each fender shall support a minimum of 300 pounds resting on any 1-square-foot area without permanent deformation. The front, rear, and outer edges of the fenders shall be rolled or angled to prevent brush damage and reduce road splash. The splash guards shall be interchangeable with guards conforming to MS51331 and shall incorporate anti-sail characteristics. The splash guards shall be of fiber glass reinforced plastic, polyester, or solid polyvinyl. The minimum physical properties of the guards shall provide an ultimate strength of 3,000 psi. When fenders are specified, instructions for the installation of the fenders shall be furnished with the skid-mounted compressor.

3.21.4 Mobility. The compressor shall be transportable while mounted on the M116 trailer chassis, as shown on MS53028, for not less than 50 miles at speeds of 25 mph over relatively smooth paved roads without any damage, twisting or distortion, or instability due to incorrect mounting. Mounting instructions including type, strength, and number of fasteners to be used shall be provided for depot mounting.

3.21.5 Crankcase drainage. Provisions shall be made to carry the compressor and engine crankcase oil drain to the outside edge of the skid base.

3.22 Type II, wheel mounted. The type II compressor shall be wheel mounted using the mounting base of the compressor as the trailer frame and using pneumatic tires adequate for the applicable load but not less than size 6.00-9. The weight of the type II compressor shall be not more than 2,350 pounds. The maximum dimensions of the type II compressor enclosure shall be 72 inches long by 60 inches wide by 46 inches high. The axle assembly shall be of the through-axle type. Axle spindles shall be heattreated and ground. The axle shall be equipped with tapered roller bearings having provisions for fine adjustment. The axle shall be perpendicular to and symetrical about the longitudinal centerline. The installation of the axle mounts shall be in accordance with the recommendations of the axle manufacturer. The axles of the trailer shall be unsprung, and the front axle, shall be trunnion mounted to compensate for the unevenness of the roadway. The knuckle steer front axle



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shall have a cramping angle of not less than 30 degrees. The thread or track of the trailer shall permit the above specified cramping angle with a clearance with the trailer frame of not less than 2 inches.

3.22.1 Brakes. Two-wheel parking brakes shall be provided. The parking brakes shall be the automotive type with internal expanding shoe, correct backing plates, and drum. The brakes shall be applied by means of a single manual control (hand lever) not exceeding 18 inches in length from the pivot point and that will simultaneously operate both brakes to lock the wheels against rotation. The lever shall be installed near the towbar where it will not be damaged or interfere with trussing of the unit for shipping. Brake actuation forces shall not exceed 200 pound-inches.

3.22.2 Towbar. The wheel mounted compressor shall have a hinged metal towbar approximately 48 inches long. The towbar shall withstand the normal loads of turning and maneuvering without deformation. The towbar on wheel mounted compressors shall be equipped with a lunette ring conforming to MS51338 and two safety chains equipped with safety-type grabhooks. Each chain shall be composed of a 1/4-inch or larger open link chain of sufficient length to reach 3 feet beyond the lunette.

3.22.3 Towing pintle. On wheel mounted compressors a towing pintle conforming to MS51335-2 shall be provided on the rear of the trailer chassis at a height from 12 to 18 inches above the ground.

3.22.4 Ground clearance. The compressor chassis shall have a ground clearance of not less than 8 inches for the pintle heights specified herein and a clearance of not less than 2 inches when ascending or descending a 20-percent slope.

3.22.5 Towing speed plate. The compressor shall be towable for 50 miles at any speed up to 25 mph over smooth paved surfaces and shall be furnished with a plate as specified in 3.16. The plate shall specify that the maximum towing speed shall be 25 mph over smooth paved surfaces. The plate shall be mounted so as to be conspicuous when the wheel-mounted compressor is being coupled for towing.

3.23 Class 1, gasoline-engine-driven.

3.23.1 Engine. For the purpose of selecting an engine, the compressor shall be classified as a class II end item in accordance with MIL-STD-1410. The engine including all systems, components, and accessories shall conform to MIL-STD-1410, class II, except as otherwise specified herein. The engine shall be furnished with a 12 volt electric cranking and battery charging system; batteries, high cylinder head temperature control of the shutdown type (if air cooled engine or high coolant shutdown if liquid cooled), low oil pressure shutdown type. All the controls shall be approved by the engine manufacturer and shall be set at the engine manufacturers recommended limits. The battery charging system shall be of the alternator type and shall be of not less than 18 ampere capacity.



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3.23.1.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the compressor under any of the operational requirements and conditions specified herein, will not require horsepower in excess of the net continuous horsepower rating of the engine for the applicable governed speed.

3.23.1.2 Air induction system. A heavy duty air cleaner and restriction indicator, in accordance with 4.2.11 of MIL-STD-1410, shall be furnished.

3.23.1.3 Exhaust system. The exhaust system shall be designed to present minumum hazard to the operator while starting, operating, or making minor adjustments to the engine or compressor unit while running. The system shall include a muffler and a spark arrester, or a combination muffler and spark arrester. The spark arrester shall conform to the US Department of Agriculture, Forest Service Standard 5100-1. The total backpressure of the exhaust system shall not exceed the maximum recommended by the engine manufacturer.

3.23.2 Fuel tank. A fuel tank with a capacity of not less than 8 hours of operation at rated load shall be provided. A filler neck assembly shall be provided and shall incorporate a removable strainer and have a chain secured filler cap and a fuel tank gasket. The filler opening shall be a minimum of 2.5 inches. Provision shall be made for connection to an auxiliary fuel supply. The fuel tank shall be equipped with a fuel measuring device. A standpipe at least 1/4 inch above the bottom shall be incorporated in the fuel tank to prevent entry of dirt and foreign matter into the system. Construction and mounting shall be such that vibration, tightening of the mounting devices, or contact with other components of the engine or compressor unit will not cause leaks to develop. The tank and all lines shall be located or positioned to minimize fire hazard or damage resulting from spilling, overflow, damage, or vapor lock at 120° F ambient temerature.

3.23.3 Clutch. A clutch shall be furnished and shall be operable from within reach of the engine controls. Provisions shall be made to prevent clutch engagement due to vibration during operation.

3.24 Class 2, electric-motor-driven.

3.24.1 Electric motor. The electric motor shall conform to CC-M-1807, type II, class 3, style E, continuous duty, totally enclosed, and as specified herein. The motor shall be 3-phase, 220/440-volt, 60-cycle, and shall be of a design which will assure low starting current and normal torque characteristics. The horsepower required by the compressors, when performing at rated capacity, shall not be more than the continuous horsepower rating, including the service factor, of the motor. All electrical equipment and wiring shall conform to the NEPA No. 70.

3.24.2 Starter switch. A pushbutton, on-off switch shall be provided.

3.24.3 Conversion items. All conversion items necessary to convert from one voltage to another shall be provided.

3.24.4 Conversion instructions. Complete instructions for converting from one voltage to another shall be permanently affixed in an accessible location.



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The conversion instructions shall be clearly readable and shall be applied to a plate conforming to 3.16.

3.25 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exhanged one for the other without alteration of the parts themselves or of adjoining parts except for adjustment, and without selection for fit or performance.

3.26 Workmanship. All parts, components, and assemblies of the compressor including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scales, or any damaging extraneous material. The compressor shall be free of any defect that could impair its operation and serviceability.

3.26.1 Fabrication. Steel and aluminum used in the fabrication of the compressors shall provide original quality surface finish and shall be free from kinks and sharp bends. The forming of the material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true, and sharp edges and burrs shall be removed. Flame cuttings may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.26.2 Welding. The surfaces of parts being welded shall be free of scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.26.3 Bolted connections. Boltholes shall be accurately punched or drilled and shall have burrs removed. Boltholes in structural members subject to variable loads shall be punched or drilled. Washers or lock washers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt.

3.26.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.26.5 Machine work. Tolerances and gages for metal fits shall conform to limitations specified herein.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Compressors which conform to all requirements in Section 3 and 5 of this specification and pass all applicable examinations and tests in Section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The contractor is responsible for assuring that components and materials used are manufactured, examined, and tested in accordance with the requirements of referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspecton (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparision (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection. Prior to examination and test of the compressor, the following shall be performed:

(a) Compressor unit power consumption. The compressor unit shall be directly coupled to a dynamometer or calibrated motor which shall be calibrated and in proper balance prior to starting the test. The compressor shall be started and operated at 3,500 psi. Determine the dynamometer speed and dynamometer load, observed brake horsepower. Inability of the compressor to deliver 15 cfm of free air at 3,500 psi in less than 1 hour after starting shall constitute failure of this test.
(b) Maintenance schedule. A schedule of maintenance to be followed during all testing of the preproduction compressor shall be furnished. A list of tools required to perform this maintenance shall also be provided.
(c) Lubrication. The compressor shall be lubricated with military oils and greases. Oils shall be those designated for use in the ambient temperature at the place of test.



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4.3.1 Examination. Prior to testing, the compressor shall be examined for the presence of defects marked "X" in column 1 of table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the compressor shall be subjected to the tests marked "X" in column 1 of table II. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production compressors will be selected at random by the Government from the compressors being produced by production tooling and will be examined for the presence of defects marked "X" in column 2 of table I and subjected to the tests marked "X" in column 2 of table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an intitial production compressor shall not exclude the remaining compressors from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Governement reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial producion tests.

4.4.1 Inspection failure. Failure of an initial production compressor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production compressor(s) and shall be cause for refusal by the Government to continue acceptance of production compressors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on compressors previously acepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all compressors accepted prior to the completion of the initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each compressor shall be examined for the presence of defects marked "X" in column 3 of table I. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each compressor shall be subjected to the tests marked "X" in column 3 of table II. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. Examination shall be in accordance with table I.



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TABLE I. Examination schedule.

| Pre-production | Initial Production | Quality Conformance | Inspection Comparison | Defects | Requirement Paragraph(s) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| X | X | X | X | Material not as specified. | 3.4 thru 3.4.2 |
| X | X | X | X | Safety devices not as specified. | 3.6.1 thru 3.6.2 |
| X | X | X | X | Human factors not as specified. | 3.6.3 |
| X | X | X | X | Maintainability not as specified. | 3.7.1 thru 3.7.2 |
| X | X | X | X | Pressure vessels, tubing and fittings not as specified. | 3.9.3 thru 3.9.4 |
| X | X | X | X | Manually operated valves not as specififed. | 3.9.5 |
| X | X | X | X | Dehydrator not as specified. | 3.9.7 thru 3.9.7.3 |
| X | X | X | X | Air filters not as specified. | 3.9.8 thru 3.9.8.3 |
| X | X | X | X | Pressure relief valves not as specified. | 3.9.9 |
| X | X | X | X | Air receiver not as specified. | 3.9.10 |
| X | X | X | X | Enclosure not as specified. | 3.9.11 |
| X | X | X | X | Winterization system and heat duct attachments not as specified, when applicable. | 3.9.12, 3.9.12.1 |
| X | X | X | X | Moisture traps and sensitive elements not as specified. | 3.9.13, 3.9.14 |
| X | X | X | X | Drain valves not as specified. | 3.9.15 |
| X | X | X | X | Drive belts not as specified. | 3.9.16 |
| X | X | X | X | Hose assemblies not as specified. | 3.9.17 |
| X | X | X | X | Servicing hose not as specified. | 3.9.18 |
| X | X | X | X | Control panel not as specified. | 3.9.19 thru 3.9.19.2 |
| X | X | X | X | Lighting not as specified. | 3.9.20 |



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TABLE I. Examination schedule (Continued).

| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| - | X | X | X | Electromagnetic interference supression components and network not identical with approved model. | 3.9.21 |
| X | X | X | X | Lubrication not as specified. | 3.9.22 thru 3.9.22.5 |
| X | X | X | X | Fungus and moisture resistance not as specified. | 3.10 |
| X | X | X | X | Lifting and tiedown attachments not as specified. | 3.11 thru 3.12 |
| X | X | X | X | Identification marking not as specified. | 3.14 |
| X | X | X | X | Shipping data plate not as specified. | 3.15 |
| X | X | X | X | Instruction plates not as specified. | 3.16 |
| X | X | X | X | Treatment and painting not as specified. | 3.17 |
| X | X | X | X | Stenciling not as specified. | 3.18 |
| X | X | X | X | Toolbox not as specified. | 3.19 |
| X | X | X | X | Skid-mounted compressor not as specified, when applicable. | 3.21 thru 3.21.5 |
| X | X | X | X | Wheel-mounted compressor not as specified, when applicable. | 3.22 thru 3.22.5 |
| X | X | X | X | Gasoline-engine-driven compressor not equipped as specified, when applicable. | 3.23.3 |
| X | X | X | X | Electric-motor-driven compressor not eqiupped as specified, when applicable. | 3.24 thru 3.24.4 |
| X | X | X | X | Interchangeability not as specified. | 3.25 |
| X | X | X | X | Workmanhip not as specified. | 3.26 thru 3.26.5 |
| - | X | X | X | Material not of same quality as used on preproduction model. | Section 3 |
| X | X | X | X | Leaks in coolant, fuel, oil, or air lines. | Section 3 |
| X | X | X | X | Cracked or broken parts or assemblies. | Section 3 |
| X | X | X | X | Parts or components missing or do not function. | Section 3 |
| X | X | X | X | Assembly incorrect or incomplete. | Section 3 |

4.6.2 Tests. Test shall be in accordance with table II.



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TABLE II. Test schedule.

| Pre-production | Initial Production | Quality Conformance | Inspection Comparison | Tests | Test Paragraph(s) | Requirement Paragraph(s) |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| X | - | - | - | Compressor unit power consumption. | 4.3(a) | 3.23.1.1, 3.24.1 |
| X | - | - | - | Maintenance schedule. | 4.3(b) | 3.7.1 |
| X | - | - | - | Lubrication. | 4.3(c) | 3.9.22.1, 6.4 |
| X | X | - | - | Test conditions. | 4.6.2.2 | 3.7.1, 3.23.1, 3.6.1 thru 3.6.3 |
| X | X | X | X | Run-in. | 4.6.2.3 | - |
| X | - | - | X | Endurance. | 4.6.2.4 | 3.8.2 |
| X | - | - | X | Final performance. | 4.6.2.4.2 | 3.8.2 |
| X | - | - | X | Operation in tilted position. | 4.6.2.5 | 3.8.3 |
| X | - | - | X | Low temperature. | 4.6.2.6.1 | 3.5.1.1 |
| X | - | - | X | High temperature. | 4.6.2.6.2 | 3.5.1, 3.5.1.2 |
| X | - | - | X | Moisture separator. | 4.6.2.7 | 3.9.6 |
| X | - | - | X | Rain. | 4.6.2.8 | 3.5.2 |
| X | - | - | X | Tiedown and lifting attachments. | 4.6.2.9 | 3.11, 3.12 |
| X | - | - | X | Rail impact. | 4.6.2.10 | 3.13 |
| X | - | - | X | Electromagnetic Interference. | 4.6.2.11 | 3.9.21 |
| X | - | - | X | Noise. | 4.6.2.12 | 3.6.4 |
| X | - | - | X | Mobility. | 4.6.2.13 | 3.21.4 |
| X | - | - | X | Brakes. | 4.6.2.13.1 | 3.22.1 |
| X | - | - | X | Shakedown. | 4.6.2.14 | 3.21.3 |
| - | X | - | - | Performance. | 4.6.2.15.1 | 3.8.1 |
| - | X | - | - | Reliability. | 4.6.2.15.2 | 3.7.3 |
| - | X | - | - | Maintainability. | 4.6.2.15.3 | 3.7.1 |

4.6.2.1 Test log. A preproduction test log covering the run-in, endurance, cycling, high temperature, low temperature, and tilted operation shall be established tabulating the appropriate test data in a similar manner to that shown in Figure 1. Storage temperature, precipitation, lifting and tiedown, pulling, towing, and rail impact test results, as applicable, shall also be recorded in this log.

4.6.2.2 Test conditions.

(a) Unless otherwise specified herein, sequence of tests is optional with the tester.



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(b) Only that maintenance established by the contractor and submitted as a maintenance schedule prior to start of testing shall be performed during the test.
(c) Safety and human factor considerations shall be evaluated throughout testing.
(d) The class 1 compressor shall be started using the electrical cranking system without external assistance.

4.6.2.3 Run-in. The compressor shall be operated for a run-in period as determined by the contractor but not less than 8 hours of operation, utilizing all operator controls. During the final hour of this test, capacity shall be determined as specified in 4.6.2.3.1. Inability of the compressor to produce the capacity or meet the performance requirements specified in 3.8.1 shall constitute failure of this test.

4.6.2.3.1 Capacity. Capacity shall be determined in accordance with the Low Pressure Nozzle Method of the ASME Performance Code for Displacement Compressors, Vacuum Pumps, and Blowers, PTC-9. Capacity may also be calculated from the weight of air delivered in a given time with a constant back pressure of 3,500 psi, wherein the pressure vessel utilized for weight measurement shall be not less than 1000 cubic inches, and capacity shall be calculated based on free air at intake conditions.

4.6.2.4 Endurance. Conduct the test without shelter and at the climatic conditions existing at the place of test. The compressor unit shall be coupled to its power unit and shall be operated for a total of 500 hours as follows:

(a) 400 hours with the compressor initially delivering not less than 15 cfm of free air at 3,500 psi.
(b) 50 hours with the compressor cycling not less than 10 times per hour between the cut-in and cutout pressure of the unloader. The output shall be adjusted so that the pressure will alternately build up to 3,500 psi, the compressor unload, and the pressure drop to the cut-in pressure.
(c) 50 hours with the compressor operating for alternate periods of 10 minutes loaded and 50 minutes unloaded.
(d) For the 400 hours of operation specified in (a), data listed in 4.6.2.4.1 shall be recorded every 2 hours, and a capacity computation shall be made after every 40 hours of operation. A graph shall be prepared, plotting capacity and corresponding air receiver discharge temperature, taken and logged every 40 hours of operation.
(e) At the end of each 50-hour cycling test specified in (b) and (c), remove the compressor from cycling operation and operate the compressor at rated capacity for 1 hour. After 1 hour operation, make a capacity computation.
(f) Capacity shall be determined utilizing the formula as set forth in 4.6.2.3.1.



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(g) Endurance test will not be credited in less than 2-hour intervals of continuous operation. Intervals of endurance test time of less than 4 hours terminated by malfunction will not be credited. All test time accrued during other successfuly completed tests specified herein shall be credited. Operate all functional controls, including valves, not less than 50 times during the test. All safety relief devices shall be caused to operate at least one time. Operation of safety devices may be demonstrated independently of compressor operation. Determine intervals of dehydrator cartridge change as specified in 3.9.7.2.

(h) Readings shall be taken only after at least 1 hour of continuous operation. Record appropriate remarks, including a record of all stops and adjustments. The amount of lubricants consumed and fuel added during the tests shall be recorded as they are added to the compressor test system.

4.6.2.4.1 Test data.

(a) Compressor crankshaft speed, in rpm.
(b) Discharge pressure measured at dehydrator outlet, in psi.
(c) Temperature of air at receiver outlet, in degrees Fahrenheit.
(d) Intercooler and aftercooler inlet and outlet surface temperatures, in degrees Fahrenheit. Thermocouples shall be used and shall be located on the cooler inlet and outlet piping as close to the cooler as physically possible.
(e) Intercooler and aftercooler pressures, in psi.
(f) Free air delivery rate, in cfm.
(g) Ambient air temperature and pressure, in degrees Fahrenheit and inches Hg.
(h) Lubricating oil temperatures, in degrees Fahrenheit.
(i) Average temperature of air inside enclosure, in degrees Fahrenheit.
(j) Dew point of delivered air, in degrees Fahrenheit.

4.6.2.4.2 Final performance. Upon completion of 500 hours of testing, repeat the capacity test specified in 4.6.2.3.1.

4.6.2.4.3 Failure criteria. Any of the following shall constitute failure of this test:

(a) Failure of any component which cannot be corrected in accordance with the contractor's recommended schedule of maintenance.
(b) Inability of the compressor to operate as specified in 3.8.1 and 3.9.2.



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(c) Inability of the valves to operate as specified in 3.9.5.
(d) A fuel tank capacity less than specified in 3.23.2.
(e) Inability of unloader to operate as specified in 3.9.10.1.
(f) Inability of the blowdown device to operate as specified in 3.9.10.2.
(g) Inability of the pressure relief valves to operate as specified in 3.9.9.
(h) A decrease in capacity of the compressor after the final performance test of more than as specified in 3.8.2.

4.6.2.5 Operation in tilted position. The compressor shall be tilted in four different positions at an angle of 15 degrees from the horizontal. In each of the different positions, start and operate the compressor for 30 minutes delivering rated capacity at 3,500 psi. Then allow the compressor to stand for 4 hours in the tilted position most likely to adversely affect the lubricaton or fuel systems, after which the compressor shall be started and operated for 30 minutes. Evidence of leaks; malfunction; faulty lubrication, cooling, or fuel supply and distribution; lubricant spill-over; or reduced performance shall constitute failure of this test.

4.6.2.6 Environmental.

4.6.2.6.1 Low Temperature. For the winterized unit, expose the compressor to an ambient temperature of Minus 50° F for no less than 4 hours or until all temperatures have been stablized for at least 1 hour. The winterization system (see 3.9.12) may be used for not more than 1 hour for heating prior to starting. For the non-winterized unit, expose the compressor to an ambient temperature of Minus 25° F for no less than 4 hours or until all temperatures have stabalized for at least 1 hour. Start the compressor unit and operate under conditions specified at full load for a minimum of 4 hours. While delivery air at 3,500 psi at a compressor crankshaft speed of not less than that used in establishing performance in 4.3(a), record the data listed in 4.6.2.4.1 at 15 minute intervals and capacity at the end of each hour. Evidence of malfunction or damage to the compressor, or inability to start and operate as specified shall consititute failure of this test.

4.6.2.6.2 High Temperature. Expose the compressor to an ambient temperature of 120° F for 4 hours or until all temperatures have stabalized for at least 1 hour while at this ambient temperature, start the compressor and operate at full load for 8 hours, delivering air at 3,500 psi at a compressor crankshaft speed of not less than that used in establishing performance in 4.3(a). Record the data listed in 4.6.2.4.1 at 30-minute intervals during operation of the compressor. Evidence of malfunction or damage to the compressor, or inability to start and operate as specified shall constitute failure of this test.

4.6.2.7 Moisture separator. Operate the compressor for a minimum of 15 minutes at a temperature of not less than 120° F and a relative humidity of not less than 85 percent. This test may be run in conjunction with the high-temperature test. Any condensed water or oil carryover from the separator into the dehydrator shall constitute failure of this test.



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4.6.2.8 Rain. Subject the compressor, in operating condition with all panels necessary for operation open, to a simulated rainstorm by spraying fresh water at a minimum rate of 3 inches per hour directed simultaneously at all four sides and on the top at an angle of approximately 45 degrees from the vertical. The apparatus for producing the spray shall emit water in the form of droplets rather than a fine mist. Expose the compressor to the spray for 1/4 hour while not running, and then start and operate the compressor for 1/4 hour at no load while the spray continues. Water leakage into air cleaners and crankcases or water accumulation on other components which prevents operation shall constitute failure of this test.

4.6.2.9 Tiedown and lifting attachments. The static load as specified in 3.11 for lifting attachments and in 3.12 for tiedown attachents shall be applied to the attachment when it is mounted on its structural member. Weld failure or permanent deformation of the attachemnt or of the structural member shall constitute failure of this test.

4.6.2.10 Rail impact. The rail impact test shall be performed prior to completion of the last 250 hours of endurance testing. The end item, in its normal shipping configuration, shall be adequately blocked and secured to the floor of the test rail car using the tie down attachments. Blocking shall be used to prevent any longitudinal, vertical or lateral movement. Loading and tie down shall be in accordance with applicable practices of the Association of American Railroads. The test car shall be impacted into two to five buffer cars. The buffer cars shall be located on a level section of track with the air and hand brakes set and the draft gear extended. The total weight of the buffer cars shall be 250,000 pounds minimum. All cars shall be equipped with Standard draft gear and conventional under frame. One impact shall take place at 10 miles per hour (mph) in one direction and one impact at 10 mph in the opposite direction. The velocities given are minimum values and should not be exceeded by more than is necessary to insure that these values are met. The speed just prior to impact shall be measured by electronic or electrical means, or both, having an accuracy of plus or minus 5 percent. The front of the compressor shall face in the direction of the empty cars. Repeat the procedure with the rear of the compressor in the direciton of the empty cars. Evidence of damage or permanent deformation to the compressor or any component or part as specified in 3.13 shall constitute failure of this test.

4.6.2.11 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.9.21. The contractor shall furnish the contracting officer the test report for evaluation. Disapproval of the report shall constitute failure of this test.

4.6.2.12 Noise. Place the compressor in a test area which is free of any large reflecting surfaces within 50 feet of the compressor or sound-level measuring equipment. The recording microphones shall be positioned at a height to simulate ear positions of operating personnel. The noise level shall be measured with the compressor operated at rated load. If the noise level is not within acceptable limits the distance from the control panel at which 85 dB(A) can be met shall be recorded.



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4.6.2.13 Mobility. Tow the type II compressor for a distance of 50 miles over smooth paved surfaces at speeds of 25 mph. Damage, twisting or distortion, instability, or inability to comply with the towing characteristics of the type II compressor shall constitute failure of this test.

4.6.2.13.1 Brakes. The type II compressor shall be towed a short distance on a paved surface free of loose material with the two-wheel parking brakes applied. Inability of the parking brakes to hold either braked wheel from rolling during towing shall constitute failure of this test.

4.6.2.14 Shakedown. The type I compressor shall be mounted on the military M-116 trailer chassis and shall be tested as specified in 4.6.2.13. Damage, twisting or distortion, or instability due to incorrect mounting of the type I compressor and fender assembly when furnished (see 3.21.3) on the M-116 trailer chassis shall constitute failure of this test.

4.6.2.15 Performance, reliability, maintainability.

4.6.2.15.1 Performance. The initial production compressor shall be operated continually through 15-minute loaded-unloaded cycles. The unloaded phase of operation shall be not more than 1 minute in duration. The test data specified in 4.6.2.4.1 shall be recorded every 2 hours during loaded operation, and a capacity computation shall be made from every fifth set of data readings. Testing shall be continued until the accumulated hours of operation and failures (if any) indicate an accept or reject decision as specified in 4.6.2.15.2.

4.6.2.15.2 Reliability. Using the MTBF specified in 3.7.3, three randomly selected compressors from the initial production lot shall be tested in accordance with 4.6.2.15.1 for a sufficient length of time to reach a decision to accept or reject in accordance with Figure 2, Test Plan. Failure for the purpose of this test is defined as any malfunction which the operator cannot remedy by adjustment, repair, or replacement within 30 minutes (all waiting periods are excluded), utilizing the resources organic to the parent unit or organization when deployed and which causes or may cause:

(a) Failure to commence operation, cessation of operation, or degradation of performance capability below the operating conditions required to continue tests specified in 4.6.2.15.1.
(b) Damage to the compressor by continued operation.
(c) Serious personnel safety hazards.

Nonconformance to 3.7.3 shall constitute failure of this test.

4.6.2.15.3 Maintainability. Maintenance operations required during initial production testing shall be accomplished in accordance with the maintenance ratio specified in 3.7.1. Nonconformance to 3.7.1 shall constitute failure of this test. During initial production testing, maintenance support elements such as manuals (when furnished as required by the contract), lubrication charts, repair parts, and special tools shall be evaluated. Inability of the compressor to be properly maintained utilizing the support elements shall constitute



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failure of this test and shall require revision of the document(s). Errors or inadequancies in the manuals shall not be considered in assessing the maintenance ratio or the mean-time-between-failure.

4.7 Inspection comparison. The Government may select compressors at any time during the contract production period and subject these compressors to the examination identified by "X" in column 4 of table I and the tests identified by "X" in column 4 of table II to determine conformance to the requirements of this specification. The inspecion will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Governemt and will not include the previously inspected preproduction model and initial production compressors. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison compressor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison compressor(s) and shall be cause for refusal by the Government to continue acceptance of production compressor until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on compressors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all compressors accepted prior to the completion of inspection comparision are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. Inspection of packaging shall be in accordance with the quality assurance providions of MIL-C-3600.

5. PACKAGING

5.1 Preservation, packing, and marking. Each complete compressor with repair parts, tools, and publications shall be preserved, packed, and marked in accordance with MIL-C-3600. The degree of preservation shall be level A, level B, or commercial and the degree of packing shall be level A, level B, or commercial as specified (see 6.2).

6. NOTES

6.1 Intended use. The compressors are intended to provide clean, dry, high-pressure air for missiles and flame throwers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specificaion.
(b) Type and class required (see 1.2).



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(c) Time frame required for submission of preproduction model and number of preproduction compressors to be furnished (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required and number of compressors to be furnished, when applicable (see 3.3).
(f) When volume capacity of receiver other than as specified is required (see 3.9.10).
(g) When winterization system is required (see 3.9.12).
(h) Composition required (see 3.10).
(i) When tow fender assemblies are required for the type I compressor (see 3.21.3).
(j) Degree of preservation and degree of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production compressors from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish compresors conforming to this specificaton.

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the compressors covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.5 Government-loaned property. For type I compressors, the contracting officer should arrange to loan the property specified in 3.20.

6.6 Definitions. For the purpose of this specification, the following definitons apply.

6.6.1 Pressure. All pressures referred to herein, unless otherwise specified as absolute or denoted by the symbol psia, shall be interpreted as psi gage.

6.6.2 Ambient temperature. Ambient temperature is defined as the temperature at atmospheric conditions surrounding the compressor but not in the immediate vicinity where temperature may be affected by the heat of the compressor or engine cooling system.

6.6.3 Free air. Free air is defined as air at atmospheric condition surrounding the compressor, at the point where the compressor is installed, but not in the immediate vicinity where it may be affected by the heat of the compressor of engine cooling system.



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6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each compressor.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of compressors. A suggested paragraph is as follows:

> "Shipment of compressors shall include repair parts, maintenance tools, operational instructions, and acessories, unless exceptions are provided elsewhere in the contract."

6.9 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4310-A085



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DATA SHEET
MFG NAME_____ P/N_____ Record Time & Amount of 1. Engine Oil Added 2. Compressor Oil Added Sheet__of__

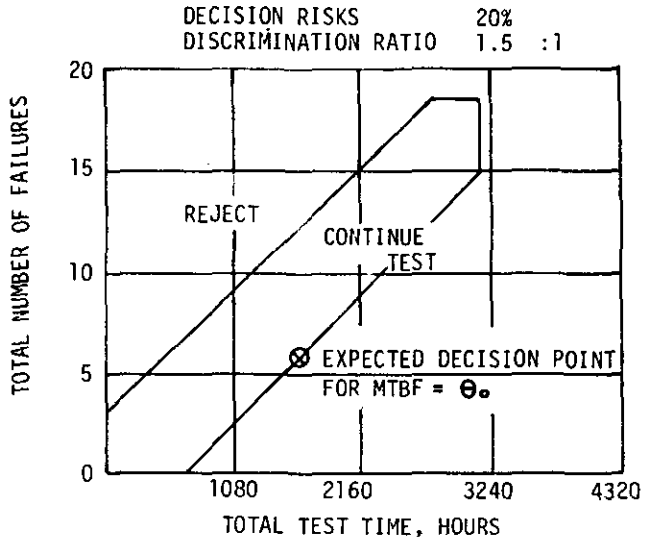
Test Title:_____ Run No._____ Date_____

| | Item | SYM | UNITS | Readings 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | ... | ... | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | Time Start, Readings, Stop | $R_t$ | HRS. MIN. | (IF STOP IS UNSCHEDULED GIVE REASON FOR STOP AND DOWN TIME) | | | | | | | | | | | | | |
| 2. | Temp. (Ambient) Dry Bulb | $T_a$ | °F | | | | | | | | | | | | | | |
| 3. | Temp. (Wet Bulb) | Twb | °F | | | | | | | | | | | | | | |
| 4. | Pressure (Ambient) | $P_b$ | In.Hg | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| 5. | Temp., Engine Coolant | $T_{e1}$ | °F | | | | | | | | | | | | | | |
| 6. | Temp., Engine Lubricant | $T_{e2}$ | °F | | | | | | | | | | | | | | |
| 7. | Temp., Compressor Lubricant | $T_L$ | °F | | | | | | | | | | | | | | |
| 8. | Temp., Receiver Discharge | $T_r$ | °F | | | | | | | | | | | | | | |
| 9. | Pressure, Rec. | $T_r$ | PSIG | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| 10. | Pressure, Compressor Intake | $P_a$ | PSIA | | | | | | | | | | | | | | |
| 11. | Temp., Compressor Intake | $T_1$ | °F | | | | | | | | | | | | | | |
| 12. | Temp., Upstream of Nozzle | $T_2$ | °F | | | | | | | | | | | | | | |
| 13. | Pres. Differential Across Nozzle | $\Delta P$ | In.$H_2O$ | | | | | | | | | | | | | | |
| 14. | Comp. Speed | N | RPM | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |

FIGURE 1. PREPRODUCTION TEST LOG X-1904

MIL-C-22072(AS)(MF)

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| NO. OF FAILURES | TOTAL TEST TIME REJECT (EQUAL OR LESS) | TOTAL TEST TIME ACCEPT (EQUAL OR LESS) | NO. OF FAILURES | TOTAL TEST TIME REJECT (EQUAL OR LESS) | TOTAL TEST TIME ACCEPT (EQUAL OR LESS) |
|---|---|---|---|---|---|
| 0 | N/A | 603 | 10 | 1261 | 2354 |
| 1 | N/A | 778 | 11 | 1436 | 2529 |
| 2 | N/A | 952 | 12 | 1611 | 2704 |
| 3 | 35 | 1128 | 13 | 1786 | 2879 |
| 4 | 210 | 1302 | 14 | 1961 | 3054 |
| 5 | 384 | 1477 | 15 | 2136 | 3154 |
| 6 | 562 | 1654 | 16 | 2311 | 3154 |
| 7 | 736 | 1830 | 17 | 2488 | 3154 |
| 8 | 912 | 2004 | 18 | 2663 | 3154 |
| 9 | 1086 | 2179 | 19 | 3154 | N/A |

Testing will continue until the total unit hours together with the total count of relevant equipment failures permit either an accept or reject decision in accordance with the specified test plan. only equipment "on" time may be used in MTBF or longetivity determinations. testing shall be monitored in such a manner that the times to failure may be estimated with reasonable accuracy.

FIGURE 2. Test Plan.

X-3478



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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-52037E(ME) Compressor Unit, Reciprocating, Power-Driven, Air, 15 CFM, 3,500-PSI

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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